

BP-Card: Structured Reporting and Validation Card for AI-Driven Cuffless Blood Pressure Estimation

A structured reporting framework for calibration-free and calibration-based approaches

MODULE 0: CLINICAL PURPOSE AND USE CONTEXT

Model name / version: _____

Estimation family: ☐ Hemodynamics-derived ☐ Data-driven BP regression ☐ ABP waveform reconstruction

Calibration paradigm: ☐ Calibration-free ☐ Calibration-based

Model type: ☐ Machine learning (ML) ☐ Deep learning (DL) ☐ Hybrid ☐ Other: _____

Input data type: ☐ Handcrafted features ☐ Time series ☐ Images ☐ Demographic ☐ Other: _____

Output variable: ☐ SBP ☐ DBP ☐ MAP

Signal modality: ☐ PPG ☐ ECG ☐ BCG ☐ BioZ ☐ iPG ☐ iPPG ☐ Other: _____

Sensor specification: _____

Sensor location: _____

Intended use: ☐ Research use only ☐ Home / ambulatory monitoring ☐ Adjunctive clinical monitoring ☐ Clinical pilot / prospective evaluation

MODULE 2. VALIDATION STRATEGY

Evaluation method: ☐ Subject-wise hold-out ☐ Subject-wise CV ☐ LOSO / LOOCV ☐ Record-wise split ☐ Sample-wise split

Data split details: _____

Unseen-subject test set: ☐ Yes ☐ No

External validation (independent cohort / site): ☐ Yes ☐ No

Reference standard: _____

Reference device (manufacturer/model): _____

Measurement conditions: ☐ Resting ☐ Ambulatory ☐ Dynamic ☐ Long-term ☐ ICU monitoring ☐ Other: _____

Body position: _____

Activity state: _____

Rest period prior to measurement: _____

Baseline comparator: _____

Comparator result: _____

Minimum acceptable standard met: _____ ☐ Subject-wise validation

Pathway-specific analyses reported separately in:

☐ Module 3A (calibration-free)

☐ Module 3B (calibration-based)

MODULE 1: DATASET AND COHORT REPORTING

Dataset source: ☐ Private ☐ Public: _____

Participants (number of subjects): _____ (at least 85 participants)

Recording sessions: _____ Total paired BP samples: _____

Population type: ☐ Healthy ☐ Hypertensive ☐ ICU ☐ Other: _____

Age (years): mean $\pm$ SD _____; range _____

Sex: % female _____; % male _____

BMI (kg/m²): mean $\pm$ SD _____

Height (cm): mean $\pm$ SD _____

Weight (kg): mean $\pm$ SD _____

Ethnicity / geographic region: _____

Medical history / comorbidities reported: ☐ Yes ☐ No

Medication reported: ☐ Yes ☐ No; if yes: class / dose / regimen _____

Known sensitivity: ☐ Motion ☐ Low perfusion ☐ Arrhythmia ☐ Sensor displacement ☐ Skin tone ☐ Other: _____

Skin tone / Fitzpatrick: _____ (if PPG-based)

Ambient light conditions: _____ (if PPG-based)

Measurement phases (%):

Ambulatory: ____ | In-clinic: ____ | Dynamic: ____ | Long-term: ____ | Sleep: ____

BP profiles (%):

Hypotension: ____ | Normal: ____ | Elevated: ____ | Stage 1 ____ | Stage 2: ____

SBP (mmHg): mean $\pm$ SD _____; range _____

DBP (mmHg): mean $\pm$ SD _____; range _____

Baseline SBP (mmHg): mean $\pm$ SD _____; range _____

Baseline DBP (mmHg): mean $\pm$ SD _____; range _____

Δ SBP (mmHg): mean $\pm$ SD ____ ☐ Subject-specific ☐ Population-based

Δ DBP (mmHg): mean $\pm$ SD ____ ☐ Subject-specific ☐ Population-based

Preprocessing pipeline: _____

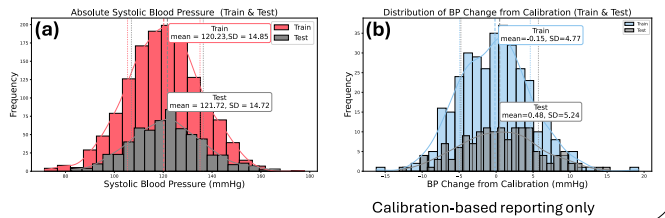
Signal quality filtering / exclusion criteria: _____

Fiducial points / feature extraction: _____

Excluded samples after cleaning: _____

BP distribution before/after cleaning reported: ☐ Yes ☐ No

BP distribution example:



MODULE 3. PERFORMANCE REPORTING

3A. Calibration-free models

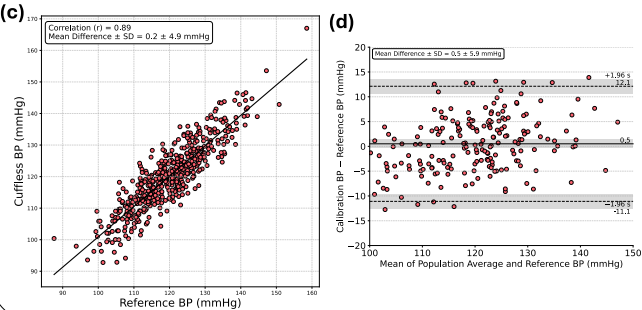
Models evaluated: ☐ Baseline model ☐ Proposed / final model

Standardized performance reporting table:

Group	Δ BP $\pm$ SD (mmHg)	MD $\pm$ SD (mmHg)	MAD (mmHg)	MAPD (%)	CP5 (%)	CP10 (%)	CP15 (%)	R
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All								
Normal								
Prehypertension								
Hypertension								
Stage 1								
Hypertension								
Stage 2								

Recommended visualization formats:



MODULE 3. PERFORMANCE REPORTING

3B. Calibration-based models

Models evaluated: ☐ Baseline model ☐ Proposed / final model

Calibration condition: _____

Calibration type: ☐ One-point ☐ Two-point ☐ Multi-point ☐ Other: _____

Static test: ☐ Yes ☐ No Dynamic test: ☐ Yes ☐ No

Time since calibration evaluated: ☐ Yes ☐ No; if yes: Time interval (days): _____

Recalibration trigger (if any): _____

Standardized performance reporting table:

Condition	Δ BP $\pm$ SD (mmHg)	MD $\pm$ SD (mmHg)	MAD (mmHg)	MAPD (%)	CP5 (%)	CP10 (%)	CP15 (%)	R
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Static								
Dynamic								
Time after calibration								

Recommended visualization formats:

